

Curriculum Vitae

1 Personal Details

- Surname: Varol
- Name: Aygün
- ORCID ID: 0000-0002-4029-7676
- Personal website: aygunvarol.github.io
- Date of CV: 29.01.2026

2 Degrees

- D.Sc. (ongoing), Computing and Electrical Engineering (Internet of Things), Tampere University
- M.Sc. (2022), Electrical and Electronics Engineering (Analog Circuit Design), Isparta University of Applied Sciences
- B.Sc. (2017), Electrical and Electronics Engineering (Industrial Automation), Manisa Celal Bayar University

3 Other Education and Expertise

- The European Summer School on Artificial Intelligence (2024), ESSAI & ACAI
- Introduction to Generative AI (2023), Google Cloud

4 Language Skills

- Native language: Turkish
- Other language skills:
 - English (Proficient - C1, IELTS: 7)
 - Finnish (Elementary)
 - German (Elementary)

5 Current Employment

- Doctoral Researcher (Since 2023), Faculty of Information Technology and Communication Sciences, Tampere University, Finland
 - My doctoral research focuses on creating intelligent indoor environments by integrating the Internet of Things (IoT) with Artificial Intelligence (AI). I am particularly interested in potential risks associated with AI deployment in these environments. With this aim, I contribute as a researcher to the EVIL-AI Project, which investigates AI-related harms and develops mitigation strategies.

6 Previous Work Experience

- Research Assistant (2020–2023), Faculty of Technology, Isparta University of Applied Sciences, Turkey
 - My main task was conducting the laboratory courses of the Fundamentals of Electrical and Electronics, Electronic, Circuit Analysis, Digital Systems, Digital System Design, System Design with Microprocessor.
- Electrical Maintenance Engineer Intern (2016), Habas Industrial and Medical Gases Production Industries Inc., Turkey
 - I was responsible for monitoring operations and maintenance in the power generation sector at a natural gas combined cycle power plant
- Electrical Maintenance Engineer Intern (2015), Çimentaş İzmir Cement Factory, Turkey
 - I was tasked with the maintenance and repair of equipment in the electrical and electronics departments.

7 Research Funding and Grants

- Study Abroad Programme Scholarship (2021), The Republic of Türkiye Ministry of National Education, monthly stipend of 1,592.50€, sponsorship period is 4 years. This scholarship programme is designed to sponsor Turkish students for post-graduate studies abroad.

8 Research Output

- Total number of publications: 7 (26 citations)
- List of publications:
 1. **Varol, A.**, Motlagh, N. H., Leino, M., Tarkoma, S., & Virkki, J. (2026). Creation of ai-driven smart spaces for enhanced indoor environments—a survey. Internet of Things, 101876.
 2. **Varol, A.**, Motlagh, N. H., Leino, M., & Virkki, J. (2025, June). Performance of Large Language Models Across Edge and Cloud Platforms in Smart Spaces. In 2025 10th International Conference on Smart and Sustainable Technologies (SpliTech) (pp. 1-6). IEEE.
 3. Shaikh, A., **Varol, A.**, & Virkki, J. (2025). From Prompts to Motors: Man-in-the-Middle Attacks on LLM-Enabled Vacuum Robots. IEEE Access.
 4. **Varol, A.**, Yucel, F., Yuce, E., & Cakir, A. (2024). A single-IC realizable, electronically tunable, OTA-based full-wave rectifier with simultaneous positive and negative outputs. AEU-International Journal of Electronics and Communications, 155374.
 5. Kayaalp, K., & **Varol, A.** (2024). LeNet ve ResNet Derin Öğrenme Modelleri ile Asma Yapraklarının Sınıflandırması. Veri Bilimi, 7(1), 16-25.
 6. **Varol, A.**, Yucel, F., & Cakir, A. (2023). A new electronically tunable transimpedance-mode OTA-based first-order universal filter and its quadrature oscillator application. Journal of Circuits, Systems and Computers, 32(11), 2350184.

7. **Varol, A., & Çetin, B. K.** (2020). A New Mobile Application For Physical Measurement in A Cellular Network. *Journal of Scientific Reports-A*, (045), 178-200.

9 Research Supervision and Leadership Experience

- 1 graduate (Master's) student co-supervision at Tampere University (2025–2026)
- 5 undergraduate (Bachelor's) research project supervisions (2020–2023), TÜBİTAK 2209-A Research Grant Program for University Students at Isparta University of Applied Sciences

10 Teaching Merits

- Teaching Assistant (2025–ongoing), The Faculty of Information Technology and Communication Sciences, Tampere University.
- Teaching Assistant (2020–2023), The Faculty of Technology, Isparta University of Applied Sciences.

11 Other Key Academic Merits

- Guest Speaker at The 1st Winter School on 3D Digitainability (2024), AI Agents in Smart Spaces, Tampere University

12 Other Merits

- **IoT & Connectivity:** End-to-end IoT prototyping (device, data collection, processing, dashboards/UIs)
- **AI/ML & LLM Engineering:** LLM fine-tuning, quantization, deployment optimization, robustness/accuracy evaluation across edge vs. cloud; both proprietary and open-source models
- **Cloud & Hybrid Inference Systems:** Edge–cloud inference workflows, API-driven services, time-series data handling for monitoring, analytics, and automation
- **Embedded & Edge Systems:** Embedded/edge deployments (e.g., Raspberry Pi sensing), performance profiling, workload optimization for constrained hardware
- **Programming Languages:** Python, C/C++, Java, $\text{\LaTeX}$
- **Tools:** Visual Studio, Android Studio, MATLAB, PSpice, Origin
- **System Design:** IoT system design/implementation; analog circuit design